

### Matrices :

$$A_{n \times m} = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1m} \\ A_{21} & A_{22} & & & \vdots \\ \vdots & & & & \\ A_{n1} & \cdots & \cdots & \cdots & A_{nm} \end{pmatrix}$$

↓  
rows      columns

### Determinant : (defined for square matrix)

1.  $|AB| = |A||B|$
2.  $|ABC \cdots| = |A||B||C| \cdots$
3.  $|k A_n| = k^n |A_n|$

### Type of matrices :

- (1) Scalar matrix  $(A = \lambda I)$
- (2) Idempotent matrix  $(A^2 = A)$
- (3) Nilpotent matrix  $(A^p = 0)$  'p is called index of A.'
- (4) Involutary matrix  $(A^2 = I)$
- (5) Periodic matrix  $(A^{k+1} = A)$  period 'k'
- (6) Symmetric matrix  $(A^T = A)$

$$\begin{array}{l} (A+B)^T = A^T + B^T \\ (kA)^T = k A^T \end{array} \quad \left| \begin{array}{l} (AB)^T = B^T A^T \\ |A^T| = |A| \end{array} \right.$$

(7) skew symmetric matrix  $(A^T = -A)$

(8) Hermitian matrix  $(A^\dagger = A)$

$$A^\dagger = ((A)^c)^T = (A^T)^c$$

– min. no. of real elements is 'n' for  $A_{n \times n}$ . ( $\bar{a}_{ii} = a_{ii} \Rightarrow$  diagonal's are real.)

$$\begin{array}{l} (A^\dagger)^\dagger = A \\ (A+B)^\dagger = A^\dagger + B^\dagger \end{array} \quad \left| \begin{array}{l} (AB)^\dagger = B^\dagger A^\dagger \\ (kA)^\dagger = \bar{k} A^\dagger \end{array} \right.$$

(9) Skew Hermitian matrix  $(A^\dagger = -A)$

$$\bar{a}_{ii} = -a_{ii} \Rightarrow a_{ii} \text{ has to be imaginary or zero.}$$

$$x - iy = -(x + iy) \Rightarrow (x = 0)$$

### Adjoint of matrix :

$$1. A(\text{adj } A) = (\text{adj } A)A = |A| I$$

$$\begin{array}{l} |A \cdot \text{adj } A| = |A|^n \\ |A| |\text{adj } A| = |A|^n \\ |\text{adj } A| = |A|^{n-1} \end{array} \quad \left| \begin{array}{l} \text{adj } A = |A| A^{-1} \\ A^{-1} = \frac{1}{|A|} \text{adj } A \end{array} \right.$$

$$2. \text{adj}(\text{adj } A) = |A|^{n-2} A.$$

$$\odot (A^T)^{-1} = (A^{-1})^T$$

$$(A^{-1})^\dagger = (A^\dagger)^{-1}$$

$$(ABC)^{-1} = C^{-1} B^{-1} A^{-1}$$

### Trace of matrix :

- $\text{Tr}(A) = \sum_{i=1}^N a_{ii}$
- $\text{Tr}(A) = \text{Tr}(A^T)$
- $\text{Tr}(AB) = \text{Tr}(BA)$
- $\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB) \hookrightarrow ABC \text{ in cyclic}$

If A, B, C are symmetric matrix

$$\text{Tr}(ABC) = \text{Tr}(BAC) = \text{Tr}(CBA) \cdots \text{All combinations.}$$

- $\text{Tr}(kA) = k \text{Tr}(A)$
- $\text{Tr}(AB) \neq \text{Tr}(A) \text{Tr}(B)$
- If  $A^T = A$  &  $B^T = -B$   $\text{Tr}(AB) = 0$
- If A is square matrix  $\boxed{|e^A| = e^{\text{Tr}(A)}}$

(10) Orthogonal matrix :

$$A^T A = I$$

$$\begin{array}{l} R_1 - \\ R_2 - \\ R_3 - \end{array} \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \quad \text{each row vector and column vector are orthonormalized.}$$

$$C_1 \quad C_2 \quad C_3$$

$$R_i R_j = \delta_{ij} \quad C_i C_j = \delta_{ij}$$

**Matrices :**

(11) unitary matrix ( $A^\dagger A = I$ )

each row & each column is orthonormalized

$$\bar{R}_i \cdot R_j = \delta_{ij} \quad |A|^2 = 1$$

$$\bar{C}_i C_j = \delta_{ij} \quad |A| = \pm 1, \frac{1+i\sqrt{3}}{2}$$

(only for diagonal matrix.)

→ Eigen values and eigen vectors :

$$A|\lambda\rangle = \lambda|\lambda\rangle$$

$$(A - \lambda I)|\lambda\rangle = 0$$

$$|\lambda\rangle = \frac{1}{|A - \lambda I|} (A - \lambda I)^{-1} 0$$

To have  $|\lambda\rangle$  ; we need  $|A - \lambda I| = 0$

→ If e.value are distinct then  $\{|\lambda_i\rangle\}$  are linearly independent & forms basis.

→ If e values are degenerate (same) then corresponding  $\{|\lambda_i\rangle\}$  may or may not be linearly independent.

vectors  $|1\rangle, |2\rangle, |3\rangle \dots$  are L.I. If

$$\sum_i a_i |i\rangle = 0 \text{ has only sol}^n$$

$$a_1 = a_2 = a_3 = a_i = 0$$

**Properties of E.values :  $e_i$**

(1)  $\lambda_i(A) = \lambda_i(A^T)$  (e.v. of  $A$  &  $A^T$  are same)

(2)  $\sum_i \lambda_i = \text{Tr}(A)$  i.e.  $\lambda_1 + \lambda_2 \dots \lambda_N = \text{Tr}(A)$

(3)  $\lambda_1 \lambda_2 \dots \lambda_N = |A|$

(4) If  $\lambda_1 \lambda_2 \lambda_3 \dots \lambda_N$  are e.v. of  $A$  then  $k\lambda_1, k\lambda_2, k\lambda_3 \dots k\lambda_N$  are e.v. of  $(kA)$

(5) If  $\lambda_1, \lambda_2, \lambda_3 \dots$  are e.v. of  $A$  then  $\lambda_1^m \lambda_2^m \dots \lambda_N^m$  are e.v. of  $A^m$

(6) If  $A$  has e.v  $\lambda_1 \lambda_2 \dots \lambda_N$  then  $A^{-1}$  has e.v:  $\lambda_1^{-1} \lambda_2^{-1} \dots \lambda_N^{-1}$  If  $(\lambda_i \neq 0)$ .

(7) If  $\begin{matrix} R_j = aR_i \\ R_k = bR_i \end{matrix}$  or  $\begin{matrix} C_2 = aC_1 \\ C_3 = bC_2 \end{matrix}$

then  $A$  has e.v. as  $(\text{Tr}(A), 0, 0, 0, 0 \dots)$

(8)  $\{\lambda_i\}$  of real symmetric matrix or Hermitian matrix are always real

(9)  $\{\lambda_i\}$  of real skew symmetrix or Skew Hermitian matrix are either zero or pure imaginary.

(10)  $|\lambda_i| = 1$  for  $A^\dagger A = I$  or  $A^T A = I$  ( $A$ : real)

(11)  $\{\lambda_i\}$  for lower/upper triangular and diagonal matrix are given by principle diagonal elements.

(12) If  $A^2 = A$  ( Idempotent matrix)  $\lambda_i = 0$  or  $1$

(13)  $(\lambda_i = 0) \forall i \leq N$  For Nilpotent matrix ( $A^m = 0$ )

**Caley–Hamilton Theorem :-**

Every square matrix satisfies its own characteristic equation.

If Characteristic eq<sup>n</sup> is quadratic

$$a\lambda^2 + b\lambda + c = 0$$

1.  $\lambda_1 + \lambda_2 = -b/a = \text{Tr}(A)$

2.  $\lambda_1 \lambda_2 = c/a = |A|$

If char. eq<sup>n</sup> is cubic:

$$a\lambda^3 + b\lambda^2 + c\lambda + d = 0$$

1.  $\lambda_1 + \lambda_2 + \lambda_3 = -b/a = \text{Tr}(A)$

2.  $\lambda_1 \lambda_2 + \lambda_2 \lambda_3 + \lambda_3 \lambda_1 = c/a$

3.  $\lambda_1 \lambda_2 \lambda_3 = d/a = |A|$ .

### Matrices :

#### Similarity transformations :

$A$  and  $B$  are said to be similar if

$$P^{-1}AP = B \quad \text{then } A = PBP^{-1}$$

If  $A$  &  $B$  are similar then,

1.  $\lambda_i$  of  $A = \lambda_i$  of  $B$

2.  $\text{Tr}(A) = \text{Tr}(B)$

3.  $|A| = |B|$  &  $\text{Rank}(A) = \text{Rank}(B)$

Matrix  $A$  can be diagonalized if  $[A, A^\dagger] = 0$

$$(P^{-1}AP = D_A) \quad \text{or } |e_i\rangle \text{ are L. Indep.}$$

To find  $P$  : For matrix  $(A)$

1. Find  $|e_i\rangle$  of  $A$

then

$$P = \begin{bmatrix} |e_1\rangle & |e_2\rangle & |e_3\rangle & \cdots \end{bmatrix}$$

If  $|e_i\rangle = |e_j\rangle$  then  $|P| = 0 \Rightarrow P^{-1}$  will not exist.

To find  $D_A$  : Find  $\lambda_i$  or  $e_i$  of matrix  $A$ .

$\cdot A, D_A$  are similar so

$$D_A = \begin{pmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \lambda_3 & \\ 0 & 0 & & \lambda_N \end{pmatrix}$$

Two matrices can be diagonalized simultaneously if they can be diagonalized with same similarity transformation matrix.

$$P^{-1}AP = D_A \quad \text{Iff } [A, B] = 0.$$

$$P^{-1}BP = D_B$$

Some properties of S.T. :

$$P^{-1}A^N P = (D_A)^N$$

$$P^{-1}e^A P = e^{D_A}$$

$$\left. \begin{aligned} \text{Tr}(\sin A) &= \sum_i \sin \lambda_i \\ |\sin A| &= \prod_{i=1}^N \sin \lambda_i \end{aligned} \right\} \text{ Same for } \cos A.$$

#### Characteristic eq<sup>n</sup> in terms of $\text{Tr}(A)$ & $|A|$ .

1.  $A_{2 \times 2}$  :  $A^2 - (\lambda_1 + \lambda_2)A + \lambda_1\lambda_2 = 0$

$$A^2 - \text{Tr} A + |A| = 0$$

2.  $A_{3 \times 3}$  :-

$$A^3 - (\lambda_1 + \lambda_2 + \lambda_3)A^2 + (\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1)A - \lambda_1\lambda_2\lambda_3 = 0$$

$$A^3 - \text{Tr} A^2 + \frac{1}{2} (\text{Tr}^2(A) - \text{Tr}(A^2)) - |A| = 0$$

If  $|A| \neq 0$  unique solution

$|A| = 0$  and  $D_1 = D_2 = D_3 = 0$  (Infinite sol<sup>n</sup>)

$|A| = 0$  and any one of  $D_i \neq 0$  (No sol<sup>n</sup>).

### Rank of matrix :

It is max no. of linearly independent vectors of a matrix. (or) order of biggest non-singular matrix.

Properties :-

(1) Rank (Null matrix) = 0

(2) Rank of non zero matrix  $\geq 1$

(3)  $R(AB) \leq R(A) \& R(B)$

#### Solution of system of linear equations :

For system of L.eq<sup>n</sup>:  $A \cdot X = B$ .

Method (1) : Find  $R(A)$  &  $R(A:B)$   $\leftarrow$  Augmented matrix

If  $R(A) = R(A:B) = n$  (unique sol<sup>n</sup>)

$R(A) = R(A:B) < n$  (Infinite sol<sup>n</sup>)

$R(A) \neq R(A:B)$  ——— (No solution)

#### Cramers Rule (Method – II) :

$$x = \frac{D_1}{|A|} \quad y = \frac{D_2}{|A|} \quad z = \frac{D_3}{|A|}$$

$$\text{For } (A \cdot X = B) \Rightarrow \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$$

$$D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} \quad D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$$

$$D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

### Differential equations :

Any eq<sup>n</sup> which consists of its variable and its derivative.

### Types :

(1) O.D.E : (Ordinary D.E) If dependent variable only depends upon one variable.

(2) P.D.E : (Partial D.E) : If dependent variable depends upon more than one variable.

Order of D.E — Highest order derivative present in D.E.

$$\partial^2 y + \partial y = \sin x \quad \text{order} \left( \frac{d^2 y}{dx^2} \right) = 2$$

Degree of D.E — Power of highest order derivative in D.E.

$$\left( \frac{d^2 y}{dx^2} \right)^2 + \left( \frac{dy}{dx} \right)^6 = 2x + 3$$

↑ (degree = 2)

General Form of D.E. →

(1) 1<sup>st</sup> order D.E — D.E whose order is 1. i.e.  $f\left(x, y, \frac{dy}{dx}\right) = 0$

(2) 2<sup>nd</sup> order D.E — D.E whose order is 2. i.e.  $f\left(x, y, \frac{dy}{dx}, \frac{d^2 y}{dx^2}\right) = 0$

(3)  $n^{th}$  order D.E — D.E whose order is ' $n$ '. i.e.  $f(x, y, y', y'', \dots y^n) = 0$

### Equations of first order and 1<sup>st</sup> degree

- (1) Variable separable    (3) Linear D.E.  
(2) Homogeneous D.E    (4) Exact D.E.

(1) variable separable :  $\frac{dy}{dx} = \frac{f(x)}{g(y)}$  or  $= \frac{f(y)}{g(x)}$

(2) Homogeneous D.E. :-

$$\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}$$

where  $f(x, y)$  &  $g(x, y)$  are of same degree in total of  $x$  &  $y$ .

$$\begin{array}{l} x \rightarrow \lambda x \\ y \rightarrow \lambda y \end{array} \quad \frac{dy}{dx} = \frac{f(\lambda x, \lambda y)}{g(\lambda x, \lambda y)} = \frac{\lambda^n f(x, y)}{\lambda^n g(x, y)}$$

$$\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}$$

Sol<sup>n</sup>: use  $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

(3) Linear D.E. If dependent variable and its differential coefficients occur in 1<sup>st</sup> degree and not multiplied together is called LDE.

e.g.  $\frac{dy}{dx} + P(x)y = Q(x)$  — 1<sup>st</sup> order LDE

2)  $x^2 \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + y = 2$  — 2<sup>nd</sup> order LDE

(Can't include terms like  $y \frac{dy}{dx}$ ,  $\left(\frac{d^2 y}{dx^2}\right)^2$ ,  $y^2 \dots$ )

Sol<sup>n</sup>:  $\frac{dy}{dx} + P(x)y = Q(x)$

$$e^{\int P dx} \frac{dy}{dx} + e^{\int P dx} P(x)y = Q(x)e^{\int P dx}$$

$$\Rightarrow \frac{d}{dx} \left( e^{\int P dx} \cdot y \right) = Q(x)e^{\int P dx}$$

$$\Rightarrow \boxed{y g = \int Q e^{\int P dx} dx}$$

Integrating factor (I.F.)

Bernoulli equation : (Reducible to linear)

$$\frac{dy}{dx} + Py = Qy^n$$

$$\Rightarrow y^{-n} \frac{dy}{dx} + Py^{1-n} = Q \quad \text{Put } y^{1-n} = t$$

$$(1-n)y^{-n} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\left( \frac{1}{1-n} \frac{dt}{dx} + Pt = Q \right) \text{ which is linear}$$

Reducible to variable separable :

$$\frac{dy}{dx} = \frac{ax + by + c}{dx + ey + h} \quad \text{Put } ax + by = t$$

$$a + b \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{1}{b} \left( \frac{dt}{dx} - a \right) = \frac{t + c}{rt + a} \leftarrow \text{variable separable}$$

### Exact D.E :

$$M(x, y)dx + N(x, y)dy = 0 \text{ is exact D.E.}$$

If  $\frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}$  then its sol<sup>n</sup> is given as

$$1. \int M_x dx + \int N(y) dy = c$$

↑ ( $x = \text{const}$ )  
only terms containing  $x$ .

$$\text{other sol}^n: \int M(x, y)dx + \int N_y dy = c$$

↑  
 $y = \text{const}$  only terms of  $N$  containing  $y$  alone.

### 2<sup>nd</sup> order L.D.E :

Eqs of type  $ay'' + by' + cy = R$

(where  $a, b, c, R$  are fun. of ' $x$ ' or Constants.)

Sol<sup>n</sup>.  $Y = C.F + P.I$

where C.F (complementary function) is sol<sup>n</sup> of  $ay'' + by' + cy = 0$

$$\text{and } P.I = \frac{1}{ay'' + by' + cy} R(x)$$

$$P.I = \frac{1}{f(D)} R(x)$$

### operator :

$$(1) \frac{d}{dx} = D, \quad \int dx = \frac{1}{D}$$

$$\text{ex, } \frac{1}{D} \cos x = \int \cos x dx = \sin x$$

$$D \cos x = \frac{d}{dx} \cos x = -\sin x.$$

### To find P.I :

#### General method :

Reduce D.E to :-  $y'' + Py' + Qy = R(x)$

$P, Q, R$  are functions of  $x$ .

$$P.I = -y_1 \int \frac{y_2 R}{W} dx + y_2 \int \frac{y_1 R}{W} dx$$

$$P.I = -y_1 \frac{1}{D} \frac{y_2 R}{W} + y_2 \frac{1}{D} \frac{y_1 R}{W}$$

where

$W = \text{Wronksian of } y_1 \text{ \& } y_2$

$y_1 \text{ \& } y_2$  are solutions of  $y'' + Py' + Qy = 0$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \text{ is called wronksian of } y_1 \text{ \& } y_2$$

$y_1 \text{ \& } y_2$  are Taken from  $(C.F = C_1 y_1 + C_2 y_2)$  of  $y'' + Py' + Qy = 0$

### Rules For finding C.F :

(1) If roots are real & different

roots of  $f(D) = 0 = m_1, m_2$

$$C.F = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

(2) If roots are equal.  $m_1 = m_2 = m$

$$C.F = (C_1 + C_2 x) e^{mx}$$

(3) If roots are imaginary.

$$m_1 = \alpha + i\beta \quad m_2 = \alpha - i\beta$$

$$Y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$= e^{\alpha x} (C_1 e^{i\beta x} + C_2 e^{-i\beta x})$$

$$= e^{\alpha x} (C_1' \cos \beta x + C_2' \sin \beta x)$$

(4) If roots are equal & imaginary.

$$m_1 = m_2 = \alpha + i\beta$$

$$Y = (C_1 + C_2 x) e^{mx}$$

$$Y = (C_1 + C_2 x) e^{(\alpha + i\beta)x}$$

$$Y = e^{\alpha x} ((C_1 + C_2 x) \cos \beta x + (C_3 + C_4 x) \sin \beta x)$$

### Equations Reducible to linear D.E.

Eq<sup>n</sup> of type :

$$(x^n D^n + x^{n-1} D^{n-1} + \dots k_n) y = Q(x) \quad \text{---(1)}$$

are reducible to linear D.E.

Take  $x = e^t \Rightarrow t = \log x \quad \left( \frac{dt}{dx} = \frac{1}{x} \right)$

$$1) \quad \frac{dy}{dx} = \frac{dy}{dt} \left( \frac{dt}{dx} \right) = \frac{1}{x} \frac{dy}{dt}$$

$$\Rightarrow x \frac{dy}{dx} = \frac{dy}{dt} = \theta y \quad \left( \theta = \frac{d}{dt} \right)$$

$$2) \quad \frac{d^2 y}{dx^2} = \frac{d}{dx} \left( \frac{dy}{dx} \right) = \frac{d}{dx} \left( \frac{1}{x} \frac{dy}{dt} \right)$$

$$= \frac{-1}{x^2} \frac{dy}{dt} + \frac{1}{x} \frac{d}{dx} \left( \frac{dy}{dt} \right) \left( \frac{dt}{dx} \right)$$

$$= \frac{-1}{x^2} \frac{dy}{dt} + \frac{1}{x} \frac{d}{dt} \left( \frac{dy}{dt} \right) \frac{dt}{dx}$$

$$\frac{d^2 y}{dx^2} = \frac{-1}{x^2} \frac{dy}{dt} + \frac{1}{x^2} \frac{d^2 y}{dt^2}$$

$$x^2 \frac{d^2 y}{dx^2} = (\theta^2 - \theta)y = \theta(\theta - 1)y$$

Similarly

$$x^3 \frac{d^3 y}{dx^3} = \theta(\theta - 1)(\theta - 2)y \quad \text{and so on.}$$

Summary: use  $x = e^t \Rightarrow t = \log x \quad \left( \frac{dt}{dx} = \frac{1}{x} \right)$

then

$$x \frac{dy}{dx} = \theta y$$

$$x^2 \frac{d^2 y}{dx^2} = \theta(\theta - 1)y \quad \left( \theta = \frac{d}{dt} \right)$$

$$x^3 \frac{d^3 y}{dx^3} = \theta(\theta - 1)(\theta - 2)y$$

use these in eq<sup>n</sup> (1) and then  $(Y = C.F + P.I)$

### Fourier series :

Any periodic function can be expressed as linear superposition of sine and cosine functions with different frequencies.

$$f(x) = a_0 + a_1 \cos x + a_2 \cos 2x + a_3 \cos 3x + \dots \\ + b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + \dots$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

(replace  $nx \longrightarrow n\omega x$ )

Integrating both sides from  $a, b$

$$\int_a^b f(x) dx = \int_a^b a_0 dx + \sum_{n=1}^{\infty} \int_a^b a_n \cos nx dx \\ + \sum_{n=1}^{\infty} \int_a^b b_n \sin(nx) dx$$

If  $f(x)$  is periodic with period  $(b - a)$ . and  $(b - a) = k(2\pi)$   
 $\hookrightarrow$  period of sine & cosine

$$a_0 = \frac{1}{(b - a)} \int_a^b f(x) dx$$

$$a_n = \frac{2}{(b - a)} \int_a^b f(x) \cos nx dx$$

$$b_n = \frac{2}{(b - a)} \int_a^b f(x) \sin nx dx$$

$b_n = 0$  If  $f(x) =$  even function

$a_n = 0$  If  $f(x) =$  odd function

$a_n = 0$  If  $\langle f(x) \rangle_a^b = 0$

Time period of  $f \pm g$  is

$$\frac{\text{LCM of } a \text{ \& } c}{\text{HCF of } b \text{ \& } d}$$

If  $f(x)$  has Time period  $a/b$

&  $g(x)$  has " "  $c/d$

### Dirichlet conditions :

(replace  $nx \rightarrow n\omega x$ )

In order to expand periodic fun. in basis of  $\cos nx$  &  $\sin nx$ , we need —

(1)  $F(x)$  should be periodic

(2)  $f(x)$  should be single valued & bounded at all points (within given interval)

( $y^2 = x$  is not single valued)

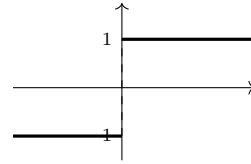
‘Bounded’ means  $f(x)$  is finite within  $(a, b)$

(3)  $F(x)$  should have finite no. of maxima or minima within one time period.

(4)  $F(x)$  should have finite number of finite discontinuity. in given interval  $[a, b]$ .

(B) If These conditions are satisfied then Fourier series converges to all points for  $f(x)$  is continuous, & converges to mean of left and right limit at point of discontinuity.

ex:



$$f(0) = \frac{f(0^+) + f(0^-)}{2} = 0$$

### Fourier Transform :

We can write any function in canonical conjugate basis set too.

$$\tilde{f}(p) = \langle p | f \rangle = \langle p | \int_{-\infty}^{\infty} dx | x \rangle \langle x | f \rangle$$

$$\tilde{f}(p) = \int_{-\infty}^{\infty} \langle p | x \rangle \langle x | f \rangle dx$$

$$\tilde{f}(p) = \int_{-\infty}^{\infty} e^{\frac{-ipx}{\hbar}} f(x) dx$$

$$\langle x | \hat{p} | p \rangle = -i\hbar \frac{\partial}{\partial x} \langle x | p \rangle$$

$$p \langle x | p \rangle = -i\hbar \frac{\partial}{\partial x} \langle x | p \rangle$$

$$\int \frac{\partial \langle x | p \rangle}{\langle x | p \rangle} = \int \frac{ip dx}{\hbar}$$

$$\left( \langle x | p \rangle = e^{\frac{ipx}{\hbar}} \right)$$

If  $\left( \frac{p}{\hbar} = k \right)$

$$\tilde{f}(p) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x) dx$$

### Inverse Fourier Transform :

$$f(x) = \langle x | f \rangle$$

$$= \int_{-\infty}^{\infty} dk \langle x | k \rangle \langle k | f \rangle$$

$$\left( f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk e^{ikx} \tilde{f}(k) \right)$$

Here  $\frac{1}{\sqrt{2\pi}}$  is a constant.

### Conditions for having Fourier Transform :

1)  $f'(x) = f(x) = 0$  at  $x = \pm\infty$

i.e.  $f(x)$  should converge

2)  $\int_{-\infty}^{\infty} f(x) dx = \text{finite} \in L_1(-\infty, \infty)$

### Central Form of gaussian function :

$$f(x) = \frac{1}{c\sqrt{2\pi}} e^{-\frac{(x-x_0)^2}{2c^2}}$$

( $c = \text{FWHM or } \Delta x$ )

$x_0 = \text{Center of gaussian fun.}$

$F(\omega)$  of Gaussian fun

$$\tilde{F} \left( \frac{1}{\sqrt{\pi c^2}} e^{-\frac{x^2}{2c^2}} \right) = \frac{1}{\sqrt{2\pi}} e^{-\frac{k^2 c^2}{2}}$$

$$c = \Delta x$$

**Laplace Transform :**

It is defined for functions which are continuous from 0 to  $\infty$ .

$$L\{f(x)\} = \int_0^{\infty} e^{-sx} f(x) dx$$

$$(1) \quad L(1) = \int_0^{\infty} e^{-sx} dx = \frac{1}{s}$$

$$(2) \quad L(e^{ax}) = \int_0^{\infty} e^{-sx} e^{ax} dx = \frac{1}{s-a}$$

$$(3) \quad L(x) = \int_0^{\infty} x e^{-sx} dx = \frac{1}{s^2}$$

(I) (II)

$$(4) \quad L(x^2) = 2/s^3$$

$$(5) \quad L(x^n) = \frac{n!}{s^{n+1}}$$

$$(6) \quad L(\cos ax) = \frac{s}{s^2 + a^2}$$

$$(7) \quad L(\sin ax) = \frac{a}{s^2 + a^2}$$

$$(8) \quad L(\cosh ax) = \frac{s}{s^2 - a^2}$$

$$(9) \quad L(\sinh ax) = \frac{a}{s^2 - a^2}$$

**Properties of Laplace Transform**

$$(1) \quad \text{If } L(f(x)) = g(s)$$

$$\text{Then, } L(e^{ax} f(x)) = g(s-a)$$

$$(2) \quad L(x^n f(x)) = (-1)^n \frac{d^n}{ds^n} g(s)$$

$$(3) \quad L(f(ax)) = \frac{1}{a} g(s/a)$$

$$\text{ex: } L(\cos^2 2x) = L\left(\frac{1 + \cos 4x}{2}\right)$$

$$= \frac{L(1)}{2} + \frac{L(\cos 4x)}{2} = \frac{1}{2s} + \frac{1}{2} \left( \frac{s}{s^2 + 4^2} \right)$$

$$(4) \quad L\left(\frac{f(x)}{x}\right) = \int_0^{\infty} g(s) ds$$

$$(5) \quad L\left(\int_0^t f(x) dx\right) = \frac{1}{s} g(s)$$

$$(6) \quad \text{Laplace Transform of periodic function with period } T.$$

$$L(f(x)) = \frac{1}{(1 - e^{-sT})} \int_0^T f(x) e^{-sx} dx$$

$$(7) \quad \text{Laplace Transform of derivatives.}$$

$$\begin{aligned} Lf^n(x) &= s^n g(s) - s^{n-1} f(x=0) \\ &\quad - s^{n-2} f'(x=0) - s^{n-3} f''(x=0) \\ &\quad \dots \end{aligned}$$

$$\text{where } g(s) = L\{f(x)\}$$

**Inverse Laplace Transform :**

$$L^{-1}(g(s)) = f(x)$$

↓

$L^{-1}$  Inverse Laplace Transform operator.

$$(\text{ex}) \quad L^{-1}\left(\frac{2}{s^3} + \frac{6}{s^2} - \frac{3}{s}\right)$$

$$= L^{-1}\left(\frac{2}{s^3}\right) + L^{-1}\left(\frac{6}{s^2}\right) - 3L^{-1}\left(\frac{1}{s}\right)$$

$$f(x) = x^2 + 6x - 3$$

$$(\text{ex}) \quad L^{-1}\left(\frac{s+3}{(s-1)(s+2)}\right) = L^{-1}\frac{1}{3}\left(\frac{4}{s-1} - \frac{1}{s+2}\right)$$

$$= \frac{(4e^t - e^{-2t})}{3}$$

$$(\text{ex}) \quad L^{-1}\left(\frac{1}{s^2 - 4s + 8}\right) = L^{-1}\left(\frac{1}{(s-2)^2 + 2^2}\right)$$

$$= \frac{1}{2} \sin 2x \cdot e^{2x}$$

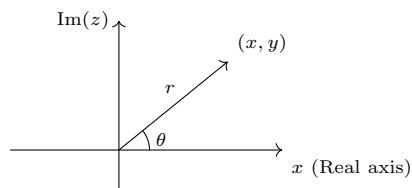


### Complex numbers

A number,  $Z = x + iy$  is represented in complex plane as shown below

$$x = \text{Re}(Z)$$

$$y = \text{Im}(Z)$$



In plane polar coordinates.

$$Z = r \cos \theta + ir \sin \theta$$

$$Z = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

$$r = \sqrt{x^2 + y^2} = \sqrt{zz^*}$$

$$\theta = \arg(z)$$

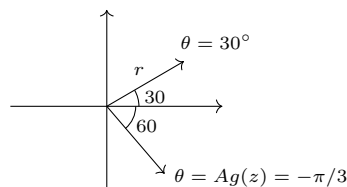
$\arg(z)$  is not unique,  $(\theta, \theta + 2\pi, \theta + 4\pi, \dots)$

whereas,  $\text{Ag}(z)$  also called as principle argument is unique.

$$-\pi \leq \text{Ag}(z) \leq \pi \quad \text{always}$$

$$\arg(z) = \text{Ag}(z) + 2n\pi \quad (n = 0, \pm 1, \pm 2, \dots)$$

$\theta = \arg(z)$  or  $\text{Ag}(z)$  is always measured from  $+x$  axis.



$\theta$ : (+ve) For measurement in anti clockwise(+)

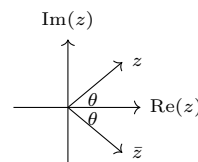
$\theta$ : (-ve) if measured in clockwise (-).

$$(1) \arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

$$(2) \arg(z_1/z_2) = \arg(z_1) - \arg(z_2)$$

$$z = x + iy = re^{i\theta}$$

$$\bar{z} = x - iy = re^{-i\theta}$$



$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$$

$$\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$$

$$\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$$

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

### $N^{th}$ roots of unity

Let  $x$  be  $n^{th}$  root of unity.

$$\text{So, } x = 1^{1/n}$$

$$1 = \cos 2m\pi + i \sin 2m\pi = e^{i2m\pi}$$

$$\text{So, } x = 1^{1/n} = e^{i\frac{2m\pi}{n}}$$

$$x = e^{i\frac{2\pi m}{n}}$$

$$m = 0, \quad x = 1 \quad (1^{\text{st}} \text{ root})$$

$$m = 1, \quad x = e^{i\frac{2\pi}{n}} = \alpha \quad (2^{\text{nd}} \text{ root})$$

$$m = 2, \quad x = e^{i\frac{4\pi}{n}} = \alpha^2$$

$$\vdots$$

$$m = n, \quad x = e^{i2\pi} = \alpha^n$$

$$S = 1 + \alpha + \alpha^2 + \dots + \alpha^{n-1} \quad (\text{Sum of } n \text{ roots of unity})$$

$$\begin{aligned} S &= \frac{1 - \alpha^n}{1 - \alpha} \quad \alpha = e^{i\frac{2\pi}{n}} \\ &= \frac{1 - e^{i2\pi}}{1 - e^{i\frac{2\pi}{n}}} = 0 \end{aligned}$$

$$\text{Similarly } P = 1 \cdot \alpha \cdot \alpha^2 \dots \alpha^{n-1}$$

$$= \alpha^{1+2+\dots+(n-1)} = \alpha^{\frac{n(n-1)}{2}}$$

$$P = e^{i\frac{2\pi}{n} \cdot \frac{n(n-1)}{2}} = e^{i\pi(n-1)}$$

$$P = \begin{cases} 1 & n \text{ even} \\ -1 & n \text{ odd} \end{cases}$$

### Complex analysis :

This is theory of functions of complex variables, in the branch of mathematical analysis that investigates functions of complex variables.

$$f(z) = u(x, y) + iv(x, y)$$

$$u = \text{Re}(f)$$

$$v = \text{Im}(f)$$

### Analytic functions :

A function  $f(z)$  is said to be analytic if  $\frac{\partial f}{\partial z^*} = 0$ .

or)  $f = u + iv$

$$u_x = v_y \quad u_y = -v_x$$

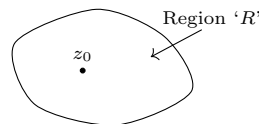
i.e.  $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$  and  $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$\hookrightarrow$  C-R relations for function to be analytic.

Analytic functions are infinitely differentiable as derivative of analytic function is also an analytic function.

### Power series representation :

We can always represent a given analytic function in some power series about some point  $z_0$ , which lies inside some region of analyticity or region of convergence.



$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \quad (\text{Taylor series})$$

where,  $a_n = \frac{1}{n!} \frac{\partial^n f}{\partial z^n} \Big|_{z=z_0}$

This power series should be convergent  $\forall z \in$  some radius of convergence.

i.e.  $\sum_{n=0}^{\infty} a_n (z - z_0)^n < \infty$

For this  $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}(z - z_0)^{n+1}}{a_n(z - z_0)^n} \right| < 1$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} (z - z_0) \right| < 1$$

$$(|z - z_0| \geq R) \quad \left\{ R < \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \right\}$$

$\uparrow$  Radius of convergence

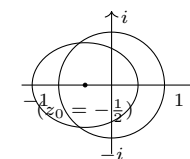
If  $R = \infty$  Then  $f(z)$  is said to be entire function.

### Example :

Write Taylor Series / power series representation of analytic function  $f(z) = 1 + z + z^2 + \dots$  about  $z_0 = -1/2$

We know that.

$$f(z) = \frac{1}{1 - z} \quad \text{for } |z| < 1.$$



So we can use  $f(z) = \frac{1}{1 - z}$  in place of  $f(z) = 1 + z + z^2 + \dots$  for  $\forall |z| < 1$ .

$$f(z) = \frac{1}{1 - z} = \frac{1}{1 - z - \frac{1}{2} + \frac{1}{2}} = \frac{1}{\frac{3}{2} - (z + \frac{1}{2})}$$

power series about  $z_0 = -1/2$  will have radius of convergence  $3/2$ , as  $f(z) = \frac{1}{1 - z}$  is singular at  $z = 1$

$$\text{So } \frac{3}{2} > |z + \frac{1}{2}|$$

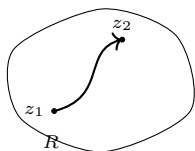
$$f(z) = \frac{1}{\frac{3}{2} (1 - \frac{2}{3} (z + \frac{1}{2}))} \quad \left[ \begin{array}{l} (1 - x)^{-1} \\ = 1 + x + x^2 + \dots \end{array} \right]$$

$$= \frac{2}{3} \left( 1 - \frac{2}{3} \left( z + \frac{1}{2} \right) \right)^{-1}$$

$$f(z) = \frac{2}{3} \left( 1 + \frac{2}{3} \left( z + \frac{1}{2} \right) + \left( \frac{2}{3} \right)^2 \left( z + \frac{1}{2} \right)^2 + \dots \right)$$

$$f(z) = \sum_{n=0}^{\infty} \left( \frac{2}{3} \right)^n \left( z + \frac{1}{2} \right)^n \quad \text{is required series.}$$

## Calculus of Residues :-

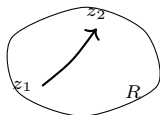


The  $\int_{z_1}^{z_2} f(z)dz$  is path independent if  $f(z)$  is analytic everywhere on  $R$ .

So As a consequence ( $I = \oint f(z)dz = 0$ )

$$\begin{aligned} I &= \int_{z_1}^{z_2} f(z)dz + \int_{z_2}^{z_1} f(z)dz \\ &= \int_{z_1}^{z_2} f(z)dz - \int_{z_1}^{z_2} f(z)dz = 0 \end{aligned}$$

$\oint f(z)dz = 0$  is called as Cauchy Integral Theorem.



$\oint f(z)dz = 0$   
If There are no singularities.

## Singularities :

### (1) Removable singularity :-

Such singularity can be removed by cleverly defining the function at singularity.

ex:  $\frac{\sin x}{x}$  has removable singularity at  $x = 0$ .

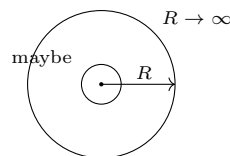
So lets redefine

$$\frac{\sin(x)}{(x)} = \begin{cases} 1 & x = 0 \\ \frac{\sin x}{x} & x \neq 0 \end{cases}$$

$$\left( \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right)$$

### (2) simple pole :-

If a function is not analytic at some point 'a' but around it it is analytic within some region of convergence.



Then we can express

$$f(z) = \frac{C_{-1}}{(z-a)} + \underbrace{\sum_{n=0}^{\infty} C_n(z-a)^n}_{\text{Regular part}}$$

$$\therefore f(z) = \underbrace{\frac{C_{-1}}{(z-a)}}_{\text{singular part}} + \underbrace{\sum_{n=0}^{\infty} C_n(z-a)^n}_{\text{Regular part}}$$

$\hookrightarrow$  simple pole

Then  $f(z)$  is said to have simple pole at  $z = a$ . with residue ( $C_{-1}$ ) at pole  $z = a$

$$\boxed{\text{Res}(z = a) = \lim_{z \rightarrow a} (z - a)f(z)}$$

$C_{(-1)}$  = Residue at  $z = a$

for  $f(z) = \frac{g(z)}{h(z)}$  If  $\begin{matrix} h(a) = 0 \\ g(a) \neq 0 \end{matrix}$

$$= \frac{g(a) + \frac{(z-a)}{1!}g'(z)|_{z=a} + \dots}{h(a) + \frac{(z-a)}{1!}h'(z)|_{z=a} + \dots}$$

$$f(z) \simeq \frac{g(a)}{h'(a)} \frac{1}{(z-a)}$$

$$\text{Residue} = \text{coeff. of } \frac{1}{z-a} = \frac{g(a)}{h'(a)}$$

### (3) Multiple poles :

$$\text{If } f(z) = \frac{C_{-m}}{(z-a)^m} + \dots + \frac{C_{-1}}{(z-a)} + \sum_{n=0}^{\infty} C_n(z-a)^n$$

$f(z)$  has Pole of order 'm' at  $z = a$

Residue is still  $C_{-1}$ .

$$\boxed{\text{Res}(f(z = a)) = \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} (z-a)^m f(z)}$$

(4) Essential / Isolated singularity.

If the singular part of  $f(z)$  contains all possible powers of  $\left(\frac{1}{z-a}\right)$  Then  $f(z)$  has isolated singularity at  $z = a$ .  
{laurent series}

$$f(z) = \underbrace{\sum_{n=1}^{\infty} \frac{C_{-n}}{(z-a)^n}}_{\text{Singular part}} + \underbrace{\sum_{n=0}^{\infty} C_n(z-a)^n}_{\text{Regular part}}$$

For convergence (singular part  $< \infty$ ) (Regular part  $< \infty$ )

$$(1) \sum_{n=1}^{\infty} \frac{C_{-n}}{(z-a)^n} < \infty \Rightarrow \text{Lt}_{n \rightarrow \infty} \left| \frac{C_{-n-1}(z-a)^n}{(z-a)^{n+1}C_{-n}} \right| < 1$$

$$\Rightarrow \text{Lt}_{n \rightarrow \infty} \left| \frac{C_{-n-1}}{C_{-n}} \frac{1}{(z-a)} \right| < 1$$

$$R_1 = |z-a| > \text{Lt}_{n \rightarrow \infty} \left| \frac{C_{-n-1}}{C_{-n}} \right|$$

$$(2) \sum_{n=0}^{\infty} C_n(z-a)^n < \infty \Rightarrow \text{Lt}_{n \rightarrow \infty} \left| \frac{C_{n+1}(z-a)}{C_n} \right| < 1$$

$$\left( R_2 = |z-a| < \text{Lt}_{n \rightarrow \infty} \left| \frac{C_n}{C_{n+1}} \right| \right)$$

Singular part is valid for

$$R_1 > \text{Lt}_{n \rightarrow \infty} \left| \frac{C_{-n-1}}{C_{-n}} \right|$$

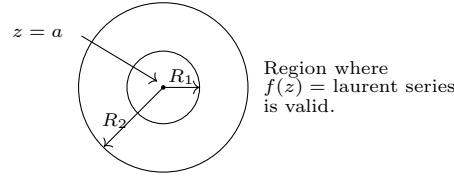
Regular part is valid for

$$R_2 < \text{Lt}_{n \rightarrow \infty} \left| \frac{C_n}{C_{n+1}} \right|$$

To have valid laurent series

$$f(z) = \sum_{n=1}^{\infty} \frac{C_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} C_n(z-a)^n$$

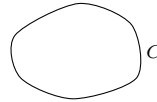
We need common region.



$$\boxed{R_1 < |z-a| < R_2} \quad (z=a) \text{ Isolated singularity}$$

Cauchy Integral Theorem :-

Consider an  $\oint_C z^n dz$



On contour  $C$  : Obviously  $z^n$  is analytic on plane

$$\text{so } I = \oint_C z^n dz = 0$$

What if  $f(z) = \frac{1}{z^{n+1}}$  for same contour.?

$$I = \oint_C \frac{1}{z^{n+1}} dz \quad \left( \begin{array}{l} f(z) \text{ has pole} \\ \text{at } z=0 \text{ of order } n+1 \end{array} \right)$$

Let  $z = e^{i\theta}$  ;  $dz = e^{i\theta} i d\theta$

$$I = \int_0^{2\pi} \frac{e^{i\theta} i d\theta}{e^{in\theta} e^{i\theta}} = i \int_0^{2\pi} e^{-in\theta} d\theta$$

$$= \frac{i (e^{-in\theta})_0^{2\pi}}{(-in)} = \frac{-1}{n} ((1-0) - (1-0)) = 0$$

$\hookrightarrow$  If  $n \neq 0$

For  $n=0$  i.e.  $f(z) = \frac{1}{z^1}$

$$\oint f(z) dz = \int \frac{1}{z} dz = \int_0^{2\pi} \frac{ie^{i\theta} d\theta}{e^{i\theta}}$$

$$\boxed{I = 2\pi i}$$

$$\text{so for } f(z) = \sum_{n=1}^{\infty} \frac{C_{-n}}{(z-a)^n} + \sum_{n=0}^{\infty} C_n(z-a)^n$$

$$\oint f(z) dz = \oint \frac{C_{-1} dz}{(z-a)} + 0$$

$$\boxed{\oint f(z) dz = (2\pi i) C_{-1}}$$

$$\boxed{\oint f(z) dz = \sum_{k=1}^{\text{all poles}} (2\pi i) \text{Res}(f(z))}$$

For  $n$  loops results will multiply by ' $n$ '.

$$\text{For Clockwise } \boxed{I = -2\pi i \text{Res}(z=a)}$$